

OpenCPU SDK

Network API Notes

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This document is specifically for **N706B** and **N520** modules.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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About This Document

Scope

This document is applicable to N706B and N520 modules.

Audience

This document is intended for [system engineers \(SEs\)](#), [development engineers](#), and [test engineers](#).

Change History

Issue	Date	Change	Changed By
1.0	2023-05	Initial release	Zhao Yating

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

1 Overview

This document introduces the Network-related APIs in the OpenCPU SDK, primarily covering the following aspects:

- Locking network modes, frequency bands, and frequency points.
- Entering and exiting flight mode.
- Retrieving network signal information.
- Obtaining information about surrounding network cells.
- Performing low-power operations such as PSM.

2 Data Structure

2.1 Enum

Due to the need to support different network information for 2G, 3G, 4G, and 5G, the data structures are numerous. It is recommended to directly refer to the .h file.

```
typedef enum {
    NWY_NW_MODE_MASK_AUTO = 0x0, //unlocked mode
    NWY_NW_MODE_MASK_CDMA_1X = 0x1,
    NWY_NW_MODE_MASK_CDMA_HDR = 0x2,
    NWY_NW_MODE_MASK_GSM = 0x4,
    NWY_NW_MODE_MASK_WCDMA = 0x8,
    NWY_NW_MODE_MASK_LTE = 0x10, //EUTRAN, Not included CATM1 and NB
    NWY_NW_MODE_MASK_TDSCDMA = 0x20,
    NWY_NW_MODE_MASK_NR = 0x40,
    NWY_NW_MODE_MASK_CATM = 0x80,
    NWY_NW_MODE_MASK_NB = 0x100,
} nwy_nw_mode_mask_e;

typedef enum {
    NWY_NW_RAT_NONE = 0, //not reg status
    NWY_NW_RAT_GSM = 1,
    NWY_NW_RAT_WCDMA = 2,
    NWY_NW_RAT_LTE = 3, //EUTRAN, Not included CATM1 and NB
    NWY_NW_RAT_CATM = 4,
    NWY_NW_RAT_NBIoT = 5,
    NWY_NW_RAT_EUTRA_TO_5GCN = 6, //LTE, core network is 5G
    NWY_NW_RAT_NR_TO_5GCN = 7, //NR, core network is 5G
    NWY_NW_RAT_NR_TO_EPS = 8, //NR, core network is 4G
    NWY_NW_RAT_EUTRA_NR_DUAL = 9, //LTE + NR
    NWY_NW_RAT_CDMA = 10,
    NWY_NW_RAT_HDR = 11,
    NWY_NW_RAT_TDSCDMA = 12,
    NWY_NW_RAT_MAX = 0xff
} nwy_nw_rat_type_e;

typedef enum {
    NWY_NW_LOCK_MODE_UNLOCK = 0, //UNLOCK all
    NWY_NW_LOCK_MODE_LOCK_BAND = 1, //lock band
    NWY_NW_LOCK_MODE_LOCK_FREQ = 2, //lock freq
    NWY_NW_LOCK_MODE_LOCK_PCI = 3 //lock cell
} nwy_nw_lockmode_e;

typedef enum {
    NWY_NW_PRI_BAND_UNLOCK = 0,
    NWY_NW_PRI_BAND_LOCK = 1,
    NWY_NW_PRI_BAND_MAX
} nwy_nw_priband_mode_e;
```

```
typedef enum {
    NWY_NW_RADIO_MIN_MODE = 0,      //min mode, close rf/sim
    NWY_NW_RADIO_NORMAL_MODE = 1,    //normal mode, open rf and sim
    NWY_NW_RADIO_FLIGHT_MODE = 4,    //flight mode, close rf
} nwy_nw_radio_mode_e;

typedef enum {
    NWY_NW_EDRX_MODE_DISABLE = 0,
    NWY_NW_EDRX_MODE_ENABLE = 1,
    NWY_NW_EDRX_MODE_ENABLE_REPORT = 2,
    NWY_NW_EDRX_MODE_RESET_SETTING = 3,
} nwy_nw_edrx_mode_e;

typedef enum {
    NWY_NW_PSM_MODE_DISABLE = 0,
    NWY_NW_PSM_MODE_ENABLE = 1,
    NWY_NW_PSM_MODE_RESET = 2,
} nwy_nw_psm_mode_e;

typedef enum {
    NWY_NW_CONFIG_RW_NETAUTO = 0,
    NWY_NW_CONFIG_RW_UEMODE = 1,
    NWY_NW_CONFIG_RW_DATAONLY = 2,
    NWY_NW_CONFIG_RW_ANTAUX = 3,
    NWY_NW_CONFIG_R_BAND_CAP = 4,
    NWY_NW_CONFIG_RW_NITZ_ONOFF = 5,
} nwy_nw_config_type_e;

typedef enum {
    NWY_NW_SPN_ENC_GSM7 = 0x00, /* 7bit code */
    NWY_NW_SPN_ENC_UCS2 = 0x01, /* UCS2 code */
} nwy_nw_spn_enc_e;

typedef enum {
    NWY_NW_SERVICE_NONE = 0, /*out of service
    NWY_NW_SERVICE_LIMITED = 1,
    NWY_NW_SERVICE_FULL = 2,
} nwy_nw_service_type_e;

typedef enum {
    NWY_NW_ROAM_STATE_OFF = 0,
    NWY_NW_ROAM_STATE_ON = 1,
} nwy_nw_roam_state_e;

typedef enum {
    NWY_BANDWIDTH_ENUM_TYPE_MIN = -1,
    NWY_LTE_BW_NRB_6_V01 = 0, /**< LTE 1.4 MHz bandwidth \n */
    NWY_LTE_BW_NRB_15_V01 = 1, /**< LTE 3 MHz bandwidth \n */
    NWY_LTE_BW_NRB_25_V01 = 2, /**< LTE 5 MHz bandwidth \n */
    NWY_LTE_BW_NRB_50_V01 = 3, /**< LTE 10 MHz bandwidth \n */
    NWY_LTE_BW_NRB_75_V01 = 4, /**< LTE 15 MHz bandwidth \n */
    NWY_LTE_BW_NRB_100_V01 = 5, /**< LTE 20 MHz bandwidth \n */
    NWY_NR5G_BW_NRB_6_V01 = 6, /**< NR5G 1.4 MHz bandwidth \n */
    NWY_NR5G_BW_NRB_15_V01 = 7, /**< NR5G 3 MHz bandwidth \n */
    NWY_NR5G_BW_NRB_25_V01 = 8, /**< NR5G 5 MHz bandwidth \n */
    NWY_NR5G_BW_NRB_50_V01 = 9, /**< NR5G 10 MHz bandwidth \n */
    NWY_NR5G_BW_NRB_75_V01 = 10, /**< NR5G 15 MHz bandwidth \n */
    NWY_NR5G_BW_NRB_100_V01 = 11, /**< NR5G 20 MHz bandwidth */
}
```

```

    NWY_BANDWIDTH_ENUM_TYPE_MAX = 2147483647
}nwy_bandwidth_type_e;

typedef enum {
    NWY_CDMA_PILOT_TYPE_ENUM_MIN_ENUM_VAL_V01 = -2147483647,
    NWY_CDMA_PILOT_CURR_ACT_PLT_V01 = 0x00,
    NWY_CDMA_PILOT_NEIGHBOR_PLT_V01 = 0x01,
    NWY_CDMA_PILOT_TYPE_ENUM_MAX_ENUM_VAL_V01 = 2147483647
}nwy_cdma_pilot_e;

typedef enum {
    NWY_CDMA_CLASS_NONE        = -1,
    NWY_CDMA_BC0                = 0,    /**< Band Class 0: U.S. Cellular band (800 MHz). */
    NWY_CDMA_BC1                = 1,    /**< Band Class 1: U.S. */
    NWY_CDMA_BC3                = 3,    /**< Band Class 3: Japanese Cellular band (800 MHz).
*/
    NWY_CDMA_BC4                = 4,    /**< Band Class 4: Korean PCS band (1800 MHz). */
    NWY_CDMA_BC5                = 5,    /**< Band Class 5 (450 MHz). */
    NWY_CDMA_BC6                = 6,    /**< Band Class 6 (2 GHz). */
    NWY_CDMA_BC7                = 7,    /**< Band Class 7 (Upper 700 MHz). */
    NWY_CDMA_BC8                = 8,    /**< Band Class 8 (1800 MHz). */
    NWY_CDMA_BC9                = 9,    /**< Band Class 9 (900 MHz). */
    NWY_CDMA_BC10               = 10,   /**< Band Class 10 (Second 800 MHz). */
    NWY_CDMA_BC11               = 11,   /**< Band Class 11: European PAMR Band (400 MHz). */
    NWY_CDMA_BC12               = 12,   /**< Band Class 12: PAMR Band (800 MHz). */
    NWY_CDMA_BC13               = 13,   /**< Band Class 13: currently undefined. */
    NWY_CDMA_BC14               = 14,   /**< Band Class 14 (U.S. PCS 1.9 GHz Band). */
    NWY_CDMA_BC15               = 15,   /**< Band Class 15 (1700-2100 MHz -AWS). */
    NWY_CDMA_BC16               = 16,   /**< Band Class 16 (U.S. 2.5 GHz). */
    NWY_CDMA_BC17               = 17,   /**< Band Class 17 (U.S. 2.5 GHz Forward Link only
band). */
    NWY_CDMA_BC18               = 18,   /**< Band Class 18 (700 MHz Public Safety Broadband).
*/
    NWY_CDMA_BC19               = 19,   /**< Band Class 19 (Lower 700 MHz band).*/

    NWY_CDMA_BC_MAX             = 20,
    /**< Upper boundary for CDMA band classes. */
}nwy_band_cdma_class_e;

typedef enum {
    NWY_GSM_NONE                = -1,

    NWY_GSM_450                 = 0,    /**< GSM 450 band (450 MHz). */
    NWY_GSM_480                 = 1,    /**< GSM 480 band (480 MHz). */
    NWY_GSM_750                 = 2,    /**< GSM 750 band (750 MHz). */
    NWY_GSM_850                 = 3,    /**< GSM 850 band (850 MHz). */
    NWY_GSM_EGSM_900            = 4,    /**< GSM Extended GSM (E-GSM) 900 band (900 MHz). */
    NWY_GSM_PGSM_900            = 5,    /**< GSM Primary GSM (P-GSM) 900 band (900 MHz). */
    NWY_GSM_RGSM_900            = 6,    /**< GSM Railways GSM (R-GSM) 900 band (900 MHz). */
    NWY_GSM_DCS_1800            = 7,    /**< GSM DCS band (1800 MHz). */
    NWY_GSM_PCS_1900            = 8,    /**< GSM PCS band (1900 MHz). */
}nwy_band_gsm_class_e;

typedef enum {
    NWY_LTE_CLASS_NONE          = 0, /* LTE bands*/

    NWY_LTE_EUTRAN_BAND1        = 1,    /**< UL:1920-1980; DL:2110-2170. */
    NWY_LTE_EUTRAN_BAND2        = 2,    /**< UL:1850-1910; DL:1930-1990. */

```



```
NWY_LTE_EUTRAN_BAND3 = 3,    /**< UL:1710-1785; DL:1805-1880. */
NWY_LTE_EUTRAN_BAND4 = 4,    /**< UL:1710-1755; DL:2110-2155. */
NWY_LTE_EUTRAN_BAND5 = 5,    /**< UL: 824-849; DL: 869- 894. */
NWY_LTE_EUTRAN_BAND6 = 6,    /**< UL: 830-840; DL: 875-885. */
NWY_LTE_EUTRAN_BAND7 = 7,    /**< UL:2500-2570; DL:2620-2690. */
NWY_LTE_EUTRAN_BAND8 = 8,    /**< UL: 880-915; DL: 925-960. */
NWY_LTE_EUTRAN_BAND9 = 9,    /**< UL:1749.9-1784.9; DL:1844.9-1879.9. */
NWY_LTE_EUTRAN_BAND10 = 10,   /**< UL:1710-1770; DL:2110-2170. */
NWY_LTE_EUTRAN_BAND11 = 11,   /**< UL:1427.9-1452.9; DL:1475.9-1500.9. */
NWY_LTE_EUTRAN_BAND12 = 12,   /**< UL:698-716; DL:728-746. */
NWY_LTE_EUTRAN_BAND13 = 13,   /**< UL: 777-787; DL: 746-756. */
NWY_LTE_EUTRAN_BAND14 = 14,   /**< UL: 788-798; DL: 758-768. */

/* Reserved for BAND 15, 16 */

NWY_LTE_EUTRAN_BAND17 = 17,   /**< UL: 704-716; DL: 734-746. */
NWY_LTE_EUTRAN_BAND18 = 18,   /**< UL: 815-830; DL: 860-875. */
NWY_LTE_EUTRAN_BAND19 = 19,   /**< UL: 830-845; DL: 875-890. */
NWY_LTE_EUTRAN_BAND20 = 20,   /**< UL: 832-862; DL: 791-821. */
NWY_LTE_EUTRAN_BAND21 = 21,   /**< UL: 1447.9-1462.9; DL: 1495.9-1510.9. */
NWY_LTE_EUTRAN_BAND23 = 23,   /**< UL: 2000-2020; DL: 2180-2200 */
NWY_LTE_EUTRAN_BAND24 = 24,   /**< UL: 1626.5-1660.5; DL: 1525 -1559. */
NWY_LTE_EUTRAN_BAND25 = 25,   /**< UL: 1850-1915; DL: 1930 -1995 . */
NWY_LTE_EUTRAN_BAND26 = 26,   /**< UL: 814-849; DL: 859 -894 . */
NWY_LTE_EUTRAN_BAND27 = 27,   /**< UL: 807.5 - 824; DL: 852 - 869 */
NWY_LTE_EUTRAN_BAND28 = 28,   /**<UL: 703-748; DL: 758-803 . */
NWY_LTE_EUTRAN_BAND29 = 29,   /**< UL: 1850-1910 or 1710 - 1755; DL: 716-728. */
NWY_LTE_EUTRAN_BAND30 = 30,   /**< UL: 2305 - 2315 ; DL: 2350 - 2360 */
NWY_LTE_EUTRAN_BAND31 = 31,   /**< UL: 452.5 - 457.5 ; DL: 462.5 - 467.5 */
NWY_LTE_EUTRAN_BAND32 = 32,   /**< DL: 9920 - 10359 */
NWY_LTE_EUTRAN_BAND33 = 33,   /**< UL: 1900-1920; DL: 1900-1920. */
NWY_LTE_EUTRAN_BAND34 = 34,   /**< UL: 2010-2025; DL: 2010-2025. */
NWY_LTE_EUTRAN_BAND35 = 35,   /**< UL: 1850-1910; DL: 1850-1910. */
NWY_LTE_EUTRAN_BAND36 = 36,   /**< UL: 1930-1990; DL: 1930-1990. */
NWY_LTE_EUTRAN_BAND37 = 37,   /**< UL: 1910-1930; DL: 1910-1930. */
NWY_LTE_EUTRAN_BAND38 = 38,   /**< UL: 2570-2620; DL: 2570-2620. */
NWY_LTE_EUTRAN_BAND39 = 39,   /**< UL: 1880-1920; DL: 1880-1920. */
NWY_LTE_EUTRAN_BAND40 = 40,   /**< UL: 2300-2400; DL: 2300-2400. */
NWY_LTE_EUTRAN_BAND41 = 41,   /**< UL: 2496-2690; DL: 2496-2690. */
NWY_LTE_EUTRAN_BAND42 = 42,   /**< UL: 3400-3600; DL: 3400-3600. */
NWY_LTE_EUTRAN_BAND43 = 43,   /**< UL: 3600-3800; DL: 3600-3800. */
NWY_LTE_EUTRAN_BAND46 = 46,   /**< DL: 5150-5925 MHz. */
NWY_LTE_EUTRAN_BAND47 = 47,   /**< DL: ? MHz. */
NWY_LTE_EUTRAN_BAND48 = 48,   /**< DL: ? MHz. */
NWY_LTE_EUTRAN_BAND49 = 49,   /**< UL: ?; DL: ? */
NWY_LTE_EUTRAN_BAND53 = 53,   /**< 2483.5 - 2495 MHz TDD */
NWY_LTE_EUTRAN_BAND59 = 59,   /**< 1785 - 1805MHz TDD */
NWY_LTE_EUTRAN_BAND67 = 67,   /**< UL: ?; DL: ? */
NWY_LTE_EUTRAN_BAND66 = 66,   /**< UL: 1710-1780; DL: 2110-2200 */
NWY_LTE_EUTRAN_BAND68 = 68,   /**< UL: ?; DL: ? */
NWY_LTE_EUTRAN_BAND71 = 71,   /**< UL: ?; DL: ? */
NWY_LTE_EUTRAN_BAND125 = 125,  /**< DL: 64835 - 64859 */
NWY_LTE_EUTRAN_BAND126 = 126,  /**< DL: 64860 - 64974 */
NWY_LTE_EUTRAN_BAND127 = 127,  /**< DL: 64975 - 64999 */
NWY_LTE_EUTRAN_BAND250 = 250,   /**< UL: 3400-3800; DL: 3400 -3800 . */
NWY_LTE_EUTRAN_BAND252 = 252,   /**< DL: 5150 - 5250 */
NWY_LTE_EUTRAN_BAND255 = 255,   /**< DL: 5725 - 5850 */
}nwy_band_lte_class_e;
```

```

typedef enum {
    NWY_WCDMA_CLASS_NONE = 0,
    NWY_WCDMA_I_IMT_2000 = 1,    /**< WCDMA Europe, Japan, and China IMT 2100 band. W1*/
    NWY_WCDMA_II_PCS_1900 = 2,   /**< WCDMA U.S. PCS 1900 band. W2*/
    NWY_WCDMA_III_1800 = 3,      /**< WCDMA Europe and China DCS 1800 band. W3*/
    NWY_WCDMA_IV_1700 = 4,       /**< WCDMA U.S. 1700 band. W4*/
    NWY_WCDMA_V_850 = 5,         /**< WCDMA U.S. 850 band. W5*/
    NWY_WCDMA_VI_800 = 6,        /**< WCDMA Japan 800 band. W6*/
    NWY_WCDMA_VII_2600 = 7,      /**< WCDMA Europe 2600 band. W7*/
    NWY_WCDMA_VIII_900 = 8,      /**< WCDMA Europe and China 900 band. W8*/
    NWY_WCDMA_IX_1700 = 9,       /**< WCDMA Japan 1700 band. W9*/
    /* Reserved W10 for WCDMA BC10-1700 band classes. */

    NWY_WCDMA_XI_1500 = 11,      /**< WCDMA 1500 band. W11*/
    NWY_WCDMA_XIX_850 = 19,      /**< WCDMA Japan 850 band. W19*/
}nwy_band_wcdma_class_e;

typedef enum {
    NWY_TDS_CLASS_NONE = 0,      /* TD-SCDMA bands */
    NWY_TDS_BANDA = 1,           /**< TDS Band A 1900-1920 MHz, 2010-2020 MHz */
    NWY_TDS_BANDB = 2,           /**< TDS Band B 1850-1910 MHz, 1930-1990 MHz */
    NWY_TDS_BANDC = 3,           /**< TDS Band C 1910-1930 MHz */
    NWY_TDS_BANDD = 4,           /**< TDS Band D 2570-2620 MHz */
    NWY_TDS_BANDE = 5,           /**< TDS Band E 2300-2400 MHz */
    NWY_TDS_BANDF = 6,           /**< TDS Band F 1880-1920 MHz */
}nwy_band_tdsdma_class_e;

typedef enum
{
    NWY_NR5G_CLASS_NONE = 0,

    /*NR bands in FR1*/
    NWY_NR5G_BAND1 = 1,          /**< UL: 1920 - 1980 MHz, DL: 2110 - 2170 MHz, FDD*/
    NWY_NR5G_BAND2 = 2,          /**< 1850 - 1910 MHz 1930 - 1990 MHz FDD */
    NWY_NR5G_BAND3 = 3,          /**< 1710 - 1785 MHz 1805 - 1880 MHz FDD */
    NWY_NR5G_BAND5 = 5,          /**< 824 - 849 MHz 869 - 894MHz FDD */
    NWY_NR5G_BAND7 = 7,          /**< 2500 - 2570 MHz 2620 - 2690 MHz FDD */
    NWY_NR5G_BAND8 = 8,          /**<880 - 915 MHz 925 - 960 MHz FDD */
    NWY_NR5G_BAND12 = 12,         /**<729 - 746 MHz 699 - 716 MHz FDD */
    NWY_NR5G_BAND13 = 13,         /**<777 - 787 MHz 746 MHz - 756 MHz FDD */
    NWY_NR5G_BAND14 = 14,         /**<758 - 768 MHz 788 - 798 MHz FDD */
    NWY_NR5G_BAND18 = 18,         /**<815 - 830 MHz 860 - 875 MHz FDD */
    NWY_NR5G_BAND20 = 20,         /**< 832 - 862 MHz 791- 821MHz FDD */
    NWY_NR5G_BAND25 = 25,         /**< 1930 - 1995 MHz 1850 - 1915 MHz FDD*/
    NWY_NR5G_BAND26 = 26,         /**<814 - 849 MHz 859 - 894 MHz FDD */
    NWY_NR5G_BAND28 = 28,         /**< 703 - 748 MHz 758 - 803 MHz FDD */
    NWY_NR5G_BAND29 = 29,         /**<N/A 717 MHz - 728 MHz SDL */
    NWY_NR5G_BAND30 = 30,         /**<2305 - 2315 MHz 2350 - 2360 MHz FDD */
    NWY_NR5G_BAND34 = 34,         /**< 2010 - 2025 MHz TDD */
    NWY_NR5G_BAND38 = 38,         /**< 2570 - 2620 MHz 2570 - 2620 MHz TDD*/
    NWY_NR5G_BAND39 = 39,         /**< 1880 - 1920 MHz TDD */
    NWY_NR5G_BAND40 = 40,         /**< 2300 - 2400 MHz TDD */
    NWY_NR5G_BAND41 = 41,         /**<2496 - 2690 MHz 2496 - 2690 MHz TDD */
    NWY_NR5G_BAND46 = 46,         /**< 5150-5925 MHz TDD */
    NWY_NR5G_BAND48 = 48,         /**< 3550 - 3700 MHz TDD */
    NWY_NR5G_BAND50 = 50,         /**< 1432 - 1517 MHz 1432 - 1517 MHz TDD */
    NWY_NR5G_BAND51 = 51,         /**< 1427 - 1432 MHz 1427 - 1432 MHz TDD */
    NWY_NR5G_BAND53 = 53,         /**< 2483.5 - 2495 MHz TDD */
}

```

```

NWY_NR5G_BAND65 = 65,    /**< 2110 - 2200 MHz    1920 - 2010 MHz    FDD    */
NWY_NR5G_BAND66 = 66,    /**< 1710 - 1780 MHz    2110 - 2200 MHz    FDD*/
NWY_NR5G_BAND70 = 70,    /**<1695 - 1710 MHz    1995- 2020 MHz    FDD */
NWY_NR5G_BAND71 = 71,    /**< 663 - 698 MHz    617 - 652 MHz    FDD */
NWY_NR5G_BAND74 = 74,    /**< 1427 -1470 MHz    1475 - 1518 MHz    FDD */
NWY_NR5G_BAND75 = 75,    /**< N/A    1432 - 1517 MHz    SDL*/
NWY_NR5G_BAND76 = 76,    /**<N/A    1427 - 1432 MHz    SDL */
NWY_NR5G_BAND77 = 77,    /**< 3.3 - 4.2 GHz    3.3 - 4.2 GHz    TDD */
NWY_NR5G_BAND78 = 78,    /**< 3.3 - 3.8 GHz    3.3 - 3.8 GHz    TDD */
NWY_NR5G_BAND79 = 79,    /**< 4.4 - 5.0 GHz    4.4 - 5.0 GHz    TDD*/
NWY_NR5G_BAND80 = 80,    /**<1710 - 1785 MHz    N/A    SUL    */
NWY_NR5G_BAND81 = 81,    /**< 880 - 915 MHz    N/A    SUL    */
NWY_NR5G_BAND82 = 82,    /**< 832 - 862 MHz    N/A    SUL    */
NWY_NR5G_BAND83 = 83,    /**< 703 - 748 MHz    N/A    SUL*/
NWY_NR5G_BAND84 = 84,    /**<1920 - 1980 MHz    N/A    SUL    */
NWY_NR5G_BAND85 = 85,    /**< 2496 - 2690 MHz    N/A    SUL    */
NWY_NR5G_BAND86 = 86,    /**< 1710 - 1780 MHz    SUL    */
NWY_NR5G_BAND91 = 91,    /**< 832 - 862 MHz    1427 - 1432 MHz    FDD */
NWY_NR5G_BAND92 = 92,    /**< 832 - 862 MHz    1432 - 1517 MHz    FDD */
NWY_NR5G_BAND93 = 93,    /**< 880 - 915 MHz    1427 - 1432 MHz    FDD */
NWY_NR5G_BAND94 = 94,    /**< 880 - 915 MHz    1432 - 1517 MHz    FDD */

/*NR bands in FR2*/
NWY_NR5G_BAND257 = 257,    /**< 26.5 -29.5 GHz    26.5 -29.5 GHz    TDD */
NWY_NR5G_BAND258 = 258,    /**< 24.25 - 27.5 GHz    24.25 - 27.5 GHz    TDD */
NWY_NR5G_BAND259 = 259,    /**< N/A    N/A    TDD */
NWY_NR5G_BAND260 = 260,    /**< 37-40 GHz    37-40 GHz    TDD */
NWY_NR5G_BAND261 = 261,    /**< 27.5-28.3 GHz    TDD */
}nwy_band_nr5g_class_e;

typedef enum {
    NWY_NW_NET_STAT_UNKNOWN    = 0,    /* Network state unknown */
    NWY_NW_NET_STAT_AVAILABLE,    /* Network is available for registration */
    NWY_NW_NET_STAT_CURRENT,    /* Network currently registered */
    NWY_NW_NET_STAT_FORBIDDEN,    /* Network forbidden registraiton */
}nwy_nw_status_e;

typedef enum {
    NWY_NW_SELECT_DEREG    = 0,
    NWY_NW_SELECT_REG_AUTO    = 1,
    NWY_NW_SELECT_REG_MANU    = 2,
    NWY_NW_SELECT_REG_MANU_AUTO    = 3, // if manual selection fails, automatic mode is
entered
}nwy_nw_select_mode_e;

typedef enum {
    NWY_NW_REGS_NONE    = -1,
    NWY_NW_REGS_DATA_IND    = 0,    // data register state event
    NWY_NW_REGS_VOICE_IND    = 1,    // voice register state event
}nwy_nw_regs_ind_type_e;

typedef enum{
    NWY_NW_CFG_DISABLE    = 0,
    NWY_NW_CFG_ENABLE    = 1,
} nwy_nw_cfg_switch_type_e;

typedef enum {
    NWY_NW_PS_MODE_II    = 0,    //UE only register EPS service,data centric

```

```

    NWY_NW_CS_PS_MODE_I = 1,    //UE registers both EPS/CS services, voice centric
    NWY_NW_CS_PS_MODI_II = 2,   //UE registers both EPS/CS services, data centric
    NWY_NW_PS_MODE_I = 3,      //UE only register EPS service,voice centric
} nwy_nw_cfg_uemode_type_e;

typedef enum{
    NWY_NW_GET_CELL = 0,        //serving cell
    NWY_NW_GET_ALL_CELL = 1,    //serving cell + neighborhood cell
} nwy_nw_cellinfo_mode_e;

```

2.2 Structures and Unions

```

typedef struct {
    int band_nums;
    uint32 band[NWY_NW_LOCK_FREQ_MAX_NUM];
}nwy_nw_lock_band_t;

typedef struct {
    int freq_nums;
    uint32 freq[NWY_NW_LOCK_FREQ_MAX_NUM];
}nwy_nw_lock_freq_t;

typedef struct {
    uint32 freq;
    int cell_nums;
    uint32 pci[NWY_NW_LOCK_PCI_MAX_NUM];
}nwy_nw_lock_pci_info_t;
typedef struct {
    int lock_pci_nums;
    nwy_nw_lock_pci_info_t lock_pci_info[NWY_NW_LOCK_PCI_MAX_NUM];
}nwy_nw_lock_pci_t;

typedef union {
    nwy_nw_lock_band_t lock_band;
    nwy_nw_lock_freq_t lock_freq;
    nwy_nw_lock_pci_t lock_pci;
}nwy_nw_lock_info_u;

typedef struct {
    nwy_nw_lockmode_e mode;
    nwy_nw_mode_mask_e rat;
    nwy_nw_lock_info_u lock_info;
} nwy_nw_lock_t;

typedef struct {
    nwy_nw_priband_mode_e mode;
    nwy_nw_mode_mask_e rat;
    int band_nums;
    int band[NWY_NW_PRI_BAND_MAX_NUM];
} nwy_nw_priband_t;

typedef struct {
    nwy_nw_edrx_mode_e mode;
    nwy_nw_mode_mask_e rat;    //radio access technology,ex.gsm/wcdma..

```

```

    uint8 requested_eDRX_value; //see Extended DRX parameters information element in 3GPP
TS 24.008 Table 10.5.5.32
    uint8 nw_provided_eDRX_value;
    uint8 Paging_time_window;
} nwy_nw_edrx_info_t;

typedef struct {
    nwy_nw_psm_mode_e mode;
    uint8 requested_Periodic_rau; //T3312, see the GPRS Timer 3 IE in 3GPP TS 24.008
Table 10.5.163a
    uint8 requested_gprs_ready_timer; //T3314, see the GPRS Timer IE in 3GPP TS 24.008
Table 10.5.172
    uint8 requested_Periodic_tau; //T3412, see the GPRS Timer 3 IE in 3GPP TS 24.008 Table
10.5.163a
    uint8 requested_active_timer; //T3324, see the GPRS Timer 2 IE in 3GPP TS 24.008 Table
10.5.163
} nwy_nw_psm_info_t;

typedef struct {
    char long_eons[NWY_NW_EONS_MAX_LEN + 1]; //long operator name,ex."CHINA MOBILE"
    char short_eons[NWY_NW_EONS_MAX_LEN + 1]; //short operator name,ex."CMCC"
    char mcc[NWY_NW_MNC_MCC_LEN + 1]; //Mobile Country Code
    char mnc[NWY_NW_MNC_MCC_LEN + 1]; //Mobile Network Code
    char spn[NWY_NW_SPN_LEN_MAX + 1]; //Service Provider Name,0xFF - Invalid
    nwy_nw_spn_enc_e spn_enc;
}nwy_nw_operator_t;

typedef struct {
    int rssi;
    int ber;
}nwy_nw_gsm_signal_info_t;

typedef struct {
    int rssi;
    int ecio;

}nwy_nw_cdma_hdr_signal_info_t;

typedef struct {
    int rscp;
    int ecio;
    int rssi;
}nwy_nw_wcdma_signal_info_t;

typedef struct {
    int rsrp;
    int rsrq;
    int rssi;
    int sinr;
}nwy_nw_lte_signal_info_t;

typedef struct {
    int rsrp;
    int rsrq;
    int rssi;
    int sinr;
}nwy_nw_nr_signal_info_t;

```

```

typedef struct {
    int ecno;
    int rscp;
    int rssi;
}nwy_nw_tdscdma_signal_info_t;

typedef union{
    nwy_nw_gsm_signal_info_t gsm_signal_info;
    nwy_nw_wcdma_signal_info_t wcdma_signal_info;
    nwy_nw_cdma_hdr_signal_info_t cdmd_hdr_signal_info;
    nwy_nw_lte_signal_info_t lte_signal_info;
    nwy_nw_nr_signal_info_t nr_signal_info;
    nwy_nw_tdscdma_signal_info_t tdscdma_signal_info;
}nwy_nw_rat_signal_info_u;

typedef struct {
    nwy_nw_rat_type_e rat;
    nwy_nw_rat_signal_info_u rat_signal_info;
}nwy_nw_signal_info_t;

typedef struct {
    nwy_nw_rat_type_e rat_type;    //Radio technology; see #Nwy_nw_radio_tech_t
    nwy_nw_service_type_e regs_state;    //Registration state
    nwy_nw_roam_state_e roam_state;
}nwy_nw_common_regs_type_t;

typedef struct {
    uint8 mm_regs_reject_info_valid; //mm:Mobile Management
    uint8 mm_regs_rej_cause;
    uint8 sm_regs_reject_info_valid; //sm:Session Management
    uint8 sm_regs_rej_cause;
    int rej_srv_domain;
}nwy_nw_res_reject_info_t;

/*The core network contains two parts, the CS domain, and the PS domain */
typedef struct {
    uint8 cs_regs_valid;
    nwy_nw_common_regs_type_t cs_regs; /* CS: 'Circuit Switching', handles voice calls in
2.3G net */
    uint8 ps_regs_valid;
    nwy_nw_common_regs_type_t ps_regs; /* PS: 'Packet Switching', handles data sessions in
2.3.4G net */
    nwy_nw_res_reject_info_t reject_info;
}nwy_nw_regstatus_t;

typedef struct {
    uint16 mcc; /*A 16-bit integer representation of MCC. Range: 0 to 999.*/
    uint16 mnc; /*A 16-bit integer representation of MNC. Range: 0 to 999.*/
    uint8 mnc_includes_pcs_digit;s
}nwy_nw_plmn_id_type_t;

typedef struct {
    nwy_cdma_pilot_e pilot_type;
    uint16 sid;
    /*System ID. Range: 0 to 32767. */
    uint16 nid;
    /* Network ID. Range: 0 to 65535.*/
    uint16 base_id;

```

```
/* Base station ID. */

nwy_nw_plmn_id_type_t plmn;
nwy_band_cdma_class_e band;
/* Active band of the current system. */
int32 channel;
/* Active channel of the current system. */
uint16 pilot_pn;
/*Pilot PN sequence offset index. Range: 0 to 511.*/
uint16 pilot_strength;
/*Strength of the pilot (in dB). Range: 0 to 64*/
uint32 base_lat;
/*Latitude of the current base station in units of 0.25 sec.*/
uint32 base_long;
/*Longitude of the current base station in units of 0.25 sec.*/
uint64 time_stamp;
/*Time (in milliseconds) from the start of GPS time when the measurement was taken.*/
int32 TxPower; /* Tx power value in 1/10 dBm resolution. */
int32 RxPower; /* Rx power value in 1/10 dBm resolution. */
}nwy_cdma_bs_info_t ;

typedef struct {
    uint32 bs_len;
    nwy_cdma_bs_info_t bs[NWY_NW_CDMA_POSITION_MAX];
}nwy_cdma_hdr_cell_info_t;

typedef struct {
    nwy_nw_plmn_id_type_t plmn;
    uint32 cell_id;
    uint32 timing_advance;
    uint16 lac; /* range 0..65535, see 24.008 10.5.1.3 */
    nwy_band_gsm_class_e band;
    uint16 arfcn; /*range 0..1023 */
    uint8 bsic; /* range 0..63, see 23.003 4.3.2 */
    uint8 egprs_support; //1: is edge
    int rssi;
    int32 TxPower; /* Tx power value in 1/10 dBm resolution. */
    int32 RxPower; /* Rx power value in 1/10 dBm resolution. */
}nwy_gsm_serv_cell_info_t;

typedef struct {
    nwy_nw_plmn_id_type_t plmn;
    uint8 bsic; /* range 0..63, see 23.003 4.3.2 */
    uint16 lac; /* range 0..65535, see 24.008 10.5.1.3 */
    uint16 cell_id;
    uint16 arfcn; /*range 0..1023 */
    int16 rssi;
    uint16 RxLev; //RX_LEV range 0..63, See 45.008 sect 8.1.4, default: 0
}nwy_nbr_gsm_cell_info_t;

typedef struct {
    nwy_nw_plmn_id_type_t plmn;
    uint16 lac;
    uint32 cellid;
    uint16 uArfcn; /*Absolute radio frequency channel number */
    uint16 psc_cellParameterId; /*Primary scrambling code*/
    uint16 set; /*Intrafrequency cell type. */
}
```

```

    int rscp;
    int ecio;
    int16 s_rxlev; /*Suitable receive level. */
}nwy_nbr_wcdma_cell_info_t;

typedef struct {
    nwy_nw_plmn_id_type_t plmn;
    uint16 tac;
    uint32 cellid;
    uint16 pci;
    uint32 earfcn;
    int rsrp;
    int rsrq;
    int rssi;
    int sinr;
    int16 srxlev; /*Suitable receive level. */
}nwy_nbr_lte_cell_info_t;

typedef struct {
    nwy_nw_plmn_id_type_t plmn;
    uint16 tac;
    uint64 nr_cellid;
    uint32 nr5g_arfcn; /**< Raster frequency of the serving cell. */
    uint16 pci; /**< Physical cell ID. */
    int rsrp;
    int rsrq;
    int rssi;
    int sinr;
    int16 srxlev; /*Suitable receive level. */
}nwy_nbr_nr_cell_info_t;

typedef struct {
    uint16 uarfcn; /*TD-SCDMA layer frequency. */
    uint16 lac;
    uint32 cellid;
    uint16 cpid; /* TDS cell ID. */
    uint16 cell_position; /*TDS cell position. */
    int16 rscp;
    int16 s_rxlev; /*Suitable receive level. */
} nwy_nbr_tdscdma_cell_info_t;

typedef struct {
    nwy_gsm_serv_cell_info_t serv_cell;

    uint32 gsm_nbr_gsm_cell_len;
    nwy_nbr_gsm_cell_info_t nbr_gsm_cell[NWY_NW_NBR_CELLS_MAX_NUM];

    uint32 gsm_nbr_wcdma_cell_len;
    nwy_nbr_wcdma_cell_info_t nbr_wcdma_cell[NWY_NW_NBR_CELLS_MAX_NUM];

    uint32 gsm_nbr_tdscdma_cell_len;
    nwy_nbr_tdscdma_cell_info_t nbr_tdscdma_cell[NWY_NW_NBR_CELLS_MAX_NUM];

    uint32 gsm_nbr_lte_cell_len;
    nwy_nbr_lte_cell_info_t nbr_lte_cell[NWY_NW_NBR_CELLS_MAX_NUM];

    uint32 gsm_nbr_nr_cell_len;
    nwy_nbr_nr_cell_info_t nbr_nr_cell[NWY_NW_NBR_CELLS_MAX_NUM];

```



```

}nwy_gsm_cell_info_t;

typedef struct {
    nwy_nw_plmn_id_type_t plmn;
    uint16 cell_id; /*Cell identity; as per 3G TS 25.331, 10.3.2.2 (28 bits) */
    uint16 lac; /*Location area code */
    uint8 is_rac_present; /*Boolean indicated is RAC is present*/
    uint8 rac;
    uint32 ura; /*URA identity */
    uint16 uarfcn; /*Absolute radio frequency channel number */
    nwy_band_wcdma_class_e band;
    uint16 psc; /*Primary scrambling code. */
    uint8 HSDPASupport; /*TRUE - serving cell supports HSDPA; FALSE - other. */
    uint8 HSUPASupport; /*TRUE - serving cell supports HSUPA; FALSE - other. \sa CCI API
Ref Manual */
    int rscp;
    int ecio;
    int rssi;
    int32 TxPower; /* Tx power value in 1/10 dBm resolution. */
    int32 RxPower; /* Rx power value in 1/10 dBm resolution. */
} nwy_wcdma_serv_cell_info_t;

typedef struct {
    nwy_wcdma_serv_cell_info_t serv_cell;

    uint8 wcdma_nbr_cell_len;
    nwy_nbr_wcdma_cell_info_t wcdma_nbr_wcdma_cell[NWY_NW_NBR_CELLS_MAX_NUM];

    uint32 wcdma_gsm_nbr_len;
    nwy_nbr_gsm_cell_info_t wcdma_nbr_gsm_cell[NWY_NW_NBR_CELLS_MAX_NUM];

    uint8 wcdma_nbr_lte_len;
    nwy_nbr_lte_cell_info_t wcdma_nbr_lte_cell[NWY_NW_NBR_CELLS_MAX_NUM];

    uint32 wcdma_nbr_nr_len;
    nwy_nbr_nr_cell_info_t wcdma_nbr_nr_cell[NWY_NW_NBR_CELLS_MAX_NUM];
}nwy_wcdma_cell_info_t;

typedef struct {
    nwy_nw_plmn_id_type_t plmn;
    uint32 cell_id; /*Cell ID (0xFFFFFFFF indicates cell ID information is not
present).*/
    uint16 lac; /*Location area code. (This field is ignored when cell_id is not
present).*/
    nwy_band_tdsdma_class_e band;
    uint16 uarfcn; /*Absolute RF channel number.*/
    uint8 cell_parameter_id; /*Cell parameter ID.*/
    uint8 pathloss; /*Path loss in units of 1 dB.*/
    uint32 timing_advance;
    int rscp;
}nwy_tdsdma_serv_cell_info_t;

typedef struct {
    nwy_nw_plmn_id_type_t plmn;
    uint16 uarfcn; /*Absolute RF channel number.*/
    uint8 cell_parameter_id; /*Cell parameter ID.*/
    int rscp;

```

```
}nwy_tdscdma_nbr_tdscdma_cell_info_t;

typedef struct {
    nwy_tdscdma_serv_cell_info_t serv_cell;
    uint8 tdscdma_nbr_cell_len;
    nwy_tdscdma_nbr_tdscdma_cell_info_t tdscdma_nbr_cell[NWY_NW_NBR_CELLS_MAX_NUM];
}nwy_tdscdma_cell_info_t;

typedef struct {
    nwy_nw_plmn_id_type_t plmn;
    uint16 tac;
    uint16 earfcn; /* EARFCN of the serving cell. Range: 0 to 65535. */
    uint32 cell_id;
    uint16 pci; /*Range: 0 to 503.*/

    uint16 band;
    uint16 dlBandwidth;
    uint8 isTdd; /* 1--Tdd, 0--Fdd */
    uint16 cqi; /*Value range: 0~15, invalid(0xFFFF). */
    int rsrp;
    int rsrq;
    int rssi;
    int sinr;
    int32 TxPower; /* Tx power value in 1/10 dBm resolution. */
    int32 RxPower; /* Rx power value in 1/10 dBm resolution. */
}nwy_lte_serv_cell_info_t;

typedef struct {
    nwy_lte_serv_cell_info_t serv_cell;

    uint8 lte_nbr_len;
    nwy_nbr_lte_cell_info_t lte_nbr_cell[NWY_NW_NBR_CELLS_MAX_NUM];

    uint8 lte_nbr_wcdma_len;
    nwy_nbr_wcdma_cell_info_t lte_nbr_wcdma_cell[NWY_NW_NBR_CELLS_MAX_NUM];

    uint8 lte_nbr_gsm_len;
    nwy_nbr_gsm_cell_info_t lte_nbr_gsm_cell[NWY_NW_NBR_CELLS_MAX_NUM];

    uint8 lte_nbr_cdma_len;
    nwy_cdma_bs_info_t lte_nbr_cdma_cell[NWY_NW_NBR_CELLS_MAX_NUM];

    uint8 lte_nbr_cdma_hdr_len;
    nwy_cdma_bs_info_t lte_nbr_cdma_hdr_cell[NWY_NW_NBR_CELLS_MAX_NUM];

    uint8 lte_nbr_nr_len;
    nwy_nbr_nr_cell_info_t lte_nbr_nr_cell[NWY_NW_NBR_CELLS_MAX_NUM];
}nwy_lte_cell_info_t;

typedef struct {
    nwy_nw_plmn_id_type_t plmn;
    uint8 tac[3]; /*R5G Serving Cell Information*/
    uint64 cell_id;
    uint32 g_node_id; // same as LTE eNodeB id,0 means invaild
    uint32 arfcn;
    nwy_band_nr5g_class_e band;
    nwy_bandwidth_type_e dlBandwidth;
```

```

uint16 pci; /*Physical cell ID. */

int rsrp;
int rsrq;
int rssi;
int sinr;
int32 TxPower; /* Tx power value in 1/10 dBm resolution. */
int32 RxPower; /* Rx power value in 1/10 dBm resolution. */
}nwy_nr_serv_cell_info_t;

typedef struct {
    nwy_nr_serv_cell_info_t serv_cell;

    uint8 nr_nbr_intra_len;
    nwy_nbr_nr_cell_info_t nr_nbr_intra_cell[NWY_NW_NBR_CELLS_MAX_NUM];

    uint8 nr_nbr_inter_len;
    nwy_nbr_nr_cell_info_t nr_nbr_inter_cell[NWY_NW_NBR_CELLS_MAX_NUM];

    uint8 nr_nbr_lte_len;
    nwy_nbr_lte_cell_info_t nr_nbr_lte_cell[NWY_NW_NBR_CELLS_MAX_NUM];

    uint8 nr_nbr_wcdma_len;
    nwy_nbr_wcdma_cell_info_t nr_nbr_wcdma_cell[NWY_NW_NBR_CELLS_MAX_NUM];
}nwy_nr5g_cell_info_t;

typedef struct
{
    nwy_nw_rat_type_e rat;
    union{
        nwy_cdma_hdr_cell_info_t cdma_hdr_cell;
        nwy_gsm_cell_info_t gsm_cell;
        nwy_wcdma_cell_info_t wcdma_cell;
        nwy_tdscdma_cell_info_t tds_cell;
        nwy_lte_cell_info_t lte_cell;
        nwy_nr5g_cell_info_t nr5g_cell;
    }cell_info;
}nwy_nw_cellinfo_t;

typedef struct {
    nwy_nw_status_e net_status;
    nwy_nw_mode_mask_e net_rat;
    nwy_nw_operator_t net_name;
}nwy_nw_net_scan_item_t;

typedef struct {
    int result;
    int net_num;
    nwy_nw_net_scan_item_t net_list[NWY_NW_SCAN_LIST_MAX_NUM];
}nwy_nw_scan_rsp_list_t;

typedef struct {
    char mcc[NWY_NW_MNC_MCC_LEN + 1];
    char mnc[NWY_NW_MNC_MCC_LEN + 1];
    nwy_nw_mode_mask_e net_rat;
    nwy_nw_select_mode_e select_mode;
}nwy_nw_select_param_t;

```

```
typedef struct{
    int8 onoff;
    int8 timer;
} nwy_nw_config_netauto_t;

typedef struct{
    uint16 support_rat;
    uint16 gsm_band;
    uint32 wcdma_band;
    uint64 lte_band[2]; //lte_band[0] for B1-B64, lte_band[1] for B65-B128
    uint64 nr_band[5]; //nr_band[0]for n1-n64;nr_band[2] for n65-n128 ...
} nwy_nw_cfg_net_capacity_t;

typedef union {
    nwy_nw_config_netauto_t netauto;
    nwy_nw_cfg_uemode_type_e uemode;
    nwy_nw_cfg_switch_type_e dataonly;
    nwy_nw_cfg_switch_type_e antaux;
    nwy_nw_cfg_net_capacity_t net_capacity;
    nwy_nw_cfg_switch_type_e nitz_onoff;
} nwy_nw_config_info_u;
```

3 APIs

Network API definitions can be found in **nwy_network.h**.

Network APIs are used to set network mode, obtain network registration status and network mode, set network mode, and monitor network status according the registered callback function of the Network API.

3.1 nwy_nw_mode_get

Function	nwy_error_e nwy_nw_mode_get(nwy_sim_id_e sim_id,int *p_mode)
Description	Retrieves the currently locked network mode.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. p_mode: Output parameter representing the network mode. Supports multiple NWY_NW_MODE_MASK configurations for various network modes. The output is in hexadecimal format, which converts to binary where each bit represents a network mode. Example: 0x4 (0000 0100): Locked to GSM only. 0x10 (0001 0000): Locked to LTE only. 0x14 (0001 0100): Locked to GSM + LTE. Refer to nwy_network.h for details on NWY_NW_MODE_MASK.
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value
Notes	• This return value does not indicate the current network registration status. • If the chip platform supports only one network mode, the response is always AUTO.

3.2 nwy_nw_mode_set

Function	nwy_error_e nwy_nw_mode_set(nwy_sim_id_e sim_id,int mode)
Description	Sets the current network mode.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. mode: Input parameter specifying the network mode. Supports multiple NWY_NW_MODE_MASK configurations for various network modes.

Return Value	Refer to nwy_nw_mode_get for details. Refer to nwy_network.h for details on NWY_NW_MODE_MASK.
	Successful: NWY_SUCCESS Failed: another enumerated value
Notes	If the chip platform supports only one network mode, the system validates the input and directly returns OK.

3.3 nwy_nw_lock_get

Function	nwy_error_e nwy_nw_lock_get(nwy_sim_id_e simid, nwy_nw_lock_t* p_lock_info)
Description	Queries the locking status for specific bands, frequencies, or cells.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. p_lock_info: Output parameter for lock information. Refer to nwy_network.h for details.
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value

3.4 nwy_nw_lock_set

Function	nwy_error_e nwy_nw_lock_set(nwy_sim_id_e simid, nwy_nw_lock_t p_lock_info)
Description	Locks specific bands, frequencies, or cells.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. p_lock_info: Input parameter for lock information. Refer to nwy_network.h for details.
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value

3.5 nwy_nw_priband_get

Function	nwy_error_e nwy_nw_priband_get(nwy_sim_id_e simid, nwy_nw_priband_t* p_priband_info)
Description	Queries the preferred band selection information.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. p_priband_info: Output parameter for preferred band information. Refer to nwy_network.h for details.
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value

3.6 nwy_nw_priband_set

Function	nwy_error_e nwy_nw_priband_set(nwy_sim_id_e simid, nwy_nw_priband_t* p_priband_info)
Description	Sets preferred band information. If network registration fails on the preferred band, it will attempt registration across all available bands.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. priband_info: Input parameter with preferred band information. Bands are sorted in descending priority in the band field of the nwy_nw_priband_t structure. Refer to nwy_network.h for details.
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value

3.7 nwy_nw_ims_get

Function	nwy_error_e nwy_nw_ims_get(nwy_sim_id_e simid, uint8* p_ims_state)
Description	Queries IMS configuration information.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. ims_state: Output parameter indicating IMS function configuration state. Refer to nwy_network.h for details.
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value
Notes	If the chip platform does not support IMS or the compilation option is not enabled, the function returns -10 (NWy_GEN_E_PLAT_NOT_SUPPORT) after validating the input.

3.8 nwy_nw_ims_set

Function	nwy_error_e nwy_nw_ims_set(nwy_sim_id_e simid, uint8 ims_state)
Description	Sets IMS configuration information.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. ims_state: Input parameter specifying the IMS function configuration state. Refer to nwy_network.h for details.
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value
Notes	If the chip platform does not support IMS or the compilation option is not enabled, the

function returns -10 (NWY_GEN_E_PLAT_NOT_SUPPORT) after validating the input.

3.9 nwy_nw_radio_get

Function	nwy_error_e nwy_nw_radio_get (int *p_radio_mode)
Description	Obtains the current Radio status
Parameter	p_radio_mode: Output parameter representing the Radio status 0: Minimum functionality (Radio and SIM power off) 1: Full functionality (default) 4: RF circuit off (Airplane mode)
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value

3.10 nwy_nw_radio_set

Function	nwy_error_e nwy_nw_radio_set (int radio_mode)
Description	Sets the current Radio status
Parameter	Radio_mode: 0: Minimum functionality (Radio and SIM power off) 1: Full functionality (default) 4: RF circuit off (Airplane mode)
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value
Notes	Some chip platforms may not differentiate between states 0 and 4, only turning off RF without powering down the SIM.

3.11 nwy_nw_edrx_get

Function	nwy_error_e nwy_nw_edrx_get (nwy_sim_id_e simid, nwy_nw_edrx_info_t *p_edrx_info)
Description	Retrieves eDRX configuration information.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. edrx_info: Output parameter with eDRX configuration details. Refer to nwy_network.h for more information.

Return Value

Successful: NWY_SUCCESS
Failed: another enumerated value

3.12 nwy_nw_edrx_set

Function

nwy_error_e nwy_nw_edrx_set (nwy_sim_id_e simid, nwy_nw_edrx_info_t * p_edrx_info)

Description

Sets eDRX configuration information.

Parameter

sim_id: Input parameter specifying the SIM card ID to operate.
edrx_info: Input parameter with eDRX configuration details. Refer to nwy_network.h for more information.

Return Value

Successful: NWY_SUCCESS
Failed: another enumerated value

3.13 nwy_nw_psm_get

Function

nwy_error_e nwy_nw_psm_get (nwy_sim_id_e simid, nwy_nw_psm_info_t * p_psm_info)

Description

Retrieves PSM configuration information.

Parameter

sim_id: Input parameter specifying the SIM card ID to operate.
psm_info: Output parameter with PSM configuration details. Refer to nwy_network.h for more information.

Return Value

Successful: NWY_SUCCESS
Failed: another enumerated value

3.14 nwy_nw_psm_set

Function

nwy_error_e nwy_nw_psm_set (int fun)

Description

Retrieves PSM configuration information.

Parameter

sim_id: Input parameter specifying the SIM card ID to operate.
psm_info: Input parameter with PSM configuration details. Refer to nwy_network.h for more information.

Return Value

Successful: NWY_SUCCESS
Failed: another enumerated value

3.15 nwy_nw_config_get

Function	nwy_error_e nwy_nw_config_get (nwy_sim_id_e simid, nwy_nw_config_type_t config_type, nwy_nw_config_info_t * p_config_info)
Description	Queries network-related configuration information.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. config_type: Input parameter indicating the type of configuration. config_info: Output parameter returning the configuration information. Refer to nwy_network.h for details.
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value

3.16 nwy_nw_config_set

Function	nwy_error_e nwy_nw_config_set (nwy_sim_id_e simid, nwy_nw_config_type_t config_type, nwy_nw_config_info_t * p_config_info)
Description	Queries network-related configuration information.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. config_type: Input parameter indicating the type of configuration. p_config_info: Input parameter providing the configuration information. Refer to nwy_network.h for details.
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value

3.17 nwy_nw_operator_get

Function	nwy_error_e nwy_nw_operator_get (nwy_sim_id_e sim_id, nwy_nw_operator_t *p_opt_name)
Description	Retrieves the operator name of the currently registered network, including Long EONS (Enhanced Operator Name String), Short EONS, MCC, and MNC. Refer to nwy_nw_operator_t for details.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. opt_name: Operator information
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value

3.18 nwy_nw_csq_get

Function	nwy_error_e nwy_nw_csq_get (nwy_sim_id_e sim_id, uint8_t *p_csq_val)
Description	Obtains current network signal strength
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. p_csq_val: Output parameter returning the current signal strength. Values are consistent with the AT command CSQ (range: 0–31; invalid: 99). (Value range is 0-31, invalid value 99)
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value

3.19 nwy_nw_signal_get

Function	nwy_error_e nwy_nw_signal_get (nwy_sim_id_e simid, nwy_nw_signal_info_t* p_signal_info)
Description	Retrieves signal-related information for the current network.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. p_signal_info: Output parameter returning signal strength information. Refer to nwy_network.h for details.
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value

3.20 nwy_nw_regstatus_get

Function	nwy_error_e nwy_nw_regstatus_get (nwy_sim_id_e simid, nwy_nw_regstatus_t *p_regs_info)
Description	Retrieves the current network registration information.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. p_regs_info: Output parameter returning registration information. Refer to nwy_network.h for details.
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value

3.21 nwy_nw_cellinfo_get

Function	nwy_error_e nwy_nw_cellinfo_get(nwy_sim_id_e simid, nwy_nw_cellinfo_mode_e cellinfo_mode, nwy_nw_cellinfo_t *p_cellinfo_info)
Description	Retrieves information about the primary and neighboring cells in the current network.
Parameter	A blocking function that can quickly return the primary cell information within 1 second. Retrieving all cell information may block for up to 5 seconds. sim_id: Input parameter specifying the SIM card ID to operate. cellinfo_mode: Specifies whether to retrieve only primary cell information or all cell information. p_cellinfo_info: Output parameter returning cell information. Refer to nwy_network.h for details.
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value
Notes	- For most platforms, cell information is retrieved from the modem's local storage after a normal network scan, without triggering a new scan. - After power-on, minimal information may be available initially due to ongoing network scans.

3.22 nwy_nw_scan

Function	nwy_error_e nwy_nw_scan(nwy_sim_id_e simid, nwy_nw_scan_cb_func scan_cb)
Description	Manually scans for networks. This is an asynchronous function, and results are returned through the callback function.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate. scan_cb: Callback function triggered upon completion of the scan, returning the scan results.
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value

3.23 nwy_nw_manual_select

Function	nwy_error_e nwy_nw_manual_select(nwy_sim_id_e simid, nwy_nw_select_param_t *p_net_select)
Description	Manually registers to a network selected from the list obtained using the nwy_nw_scan function.
Parameter	sim_id: Input parameter specifying the SIM card ID to operate.

	p_net_select: Input parameter specifying the network selection parameters, including the PLMN and radio access technology.
Return Value	Successful: NWY_SUCCESS Failed: another enumerated value